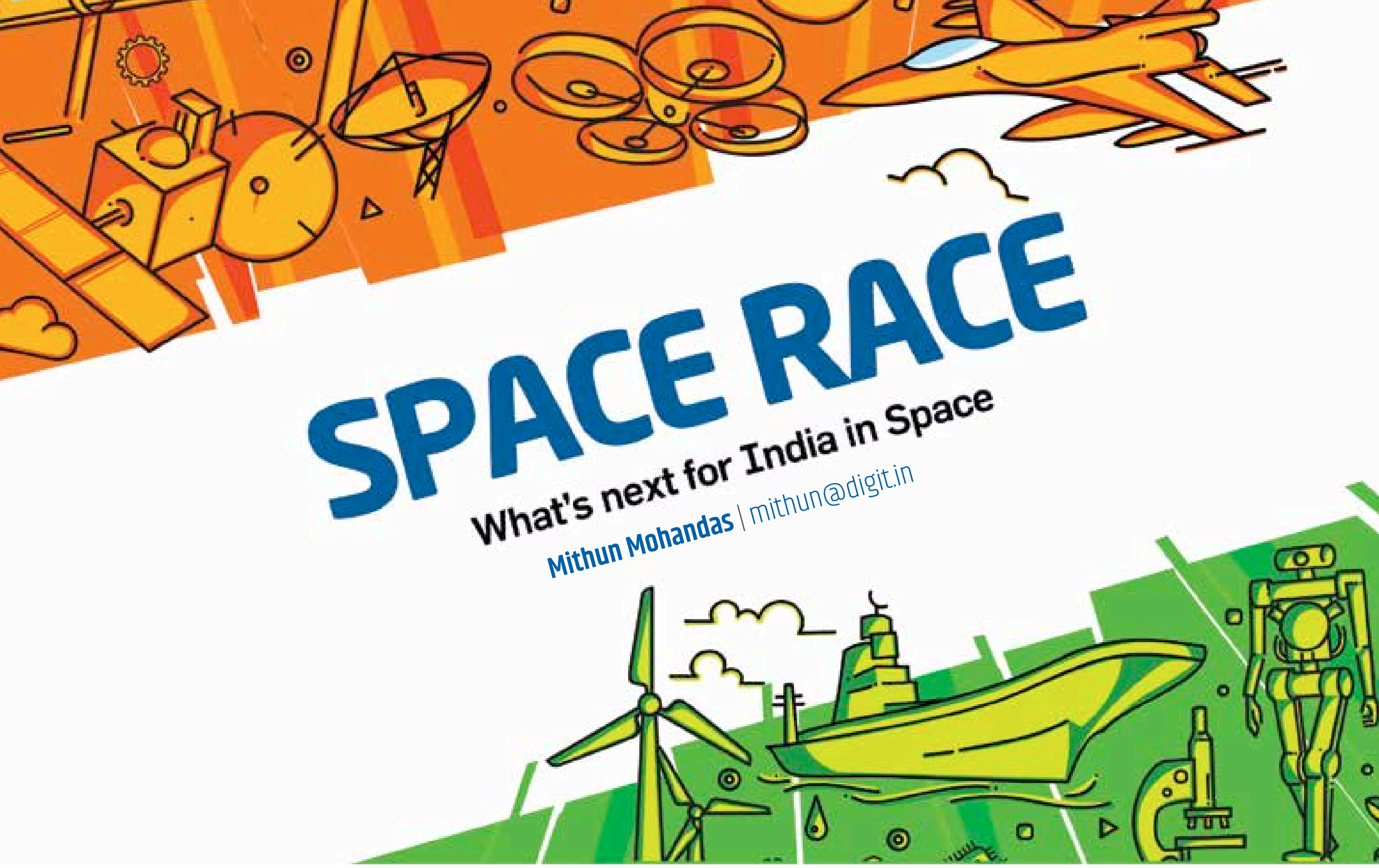

SPACE RACE

What's next for India in Space

Mithun Mohandas | mithun@digit.in

The Indian Space Research Organisation has been the underdog amongst the world's space agencies. And just like all the hundreds of movies wherein the underdog makes its way to the top, our very own ISRO is charting its own path to the stars. At any given time, ISRO is working on numerous projects which range from building satellites for communication, navigation, research, etc. to designing a Reusable Launch Vehicle that might just give SpaceX's Falcon 9 a run for its money. We're going to be looking at some of these projects which are truly pushing the boundaries of innovation.

CHANDRAYAAN-II

ISRO's first lunar mission i.e. Chandrayaan-1 was launched back in 2008 and made India the fourth country to ever place its flag on the moon. Chandrayaan-II is a much more complicated and multi-tiered mission as compared to its predecessor. It will have a Lander and a Rover which will be launched onto the surface of the moon by the orbiter. As was with the previous mission, Chandrayaan-II will be launched into space and will enter Earth Parking Orbit before gaining speed and being catapulted across to the moon.

Once the orbiter enters the Lunar orbit, the Lander onto which the Rover is attached will separate from the orbiter and it will perform a soft landing on the moon. Once on the surface, the Rover will be deployed and it'll go about doing the usual deeds that Rovers do i.e. perform mineralogical studies, take plenty of photos and run a couple of experiments.

Chandrayaan-II is scheduled to be launched in 2018 and will use the GSLV-Mk II launch vehicle. While GSLV-Mk III has had two successful flights so far, it's preferred to let the older and more mature platform for such a prestigious mission.

RLV-TD

The Reusable Launch Vehicle - Technology Demonstration Programme is a series of missions being planned by ISRO towards making a Two Stage to Orbit (TSTO) fully Reusable Vehicle. If that sounds a little too complicated then allow us to use a far more familiar example of SpaceX's Falcon 9 which is also a TSTO reusable vehicle. While we are a couple of years away from getting the RLV-TD fully functional, we are frequently executing different phases of the RLV-TD.

These phases or experiments involve testing different aspects of the RLV-TD. The first two of these series

(Credits: ISRO)

Reusable Launch Vehicle - Technology Demonstration Program (RLV-TD)

of experiments were launched in 2016, namely, the HEX (Hypersonic Flight Experiment) and the SPEX (Scramjet Propulsion Experiment). Two more experiments, the LEX (Landing Experiment) and the REX (Return Flight Experiment) will be announced soon. Once these four experiments have been successfully completed, the individual components of the RLV-TD will have proven their worth. All that remains after that is the HEX1 (Hypersonic Ex-